Self-similar solution [edit]

The gas motion behind the shock wave is governed by Euler equations. For an ideal polytropic gas with spherical symmetry, the equations for the fluid variables such as radial velocity $v(r,t)$, density $\rho(r,t)$ and pressure $p(r,t)$ are given by

$$\frac{\partial v}{\partial t} + v\frac{\partial v}{\partial r} = -\frac{1}{\rho}\frac{\partial p}{\partial r},$$
$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho v)}{\partial r} = -\frac{2\rho v}{r},$$
$$\left(\frac{\partial}{\partial t} + v\frac{\partial}{\partial r}\right)\ln\frac{p}{\rho^\gamma} = 0.$$

At $r = R(t)$, the solutions should approach the values given by the Rankine-Hugoniot conditions defined in the previous section.

The variable pressure can be replaced by the sound speed $c(r,t)$ since pressure can be obtained from the formula $c^2 = \gamma p/\rho$. The following non-dimensional self-similar variables are introduced,[13][14]

$$\xi = \frac{r}{R(t)}, \quad V(\xi) = \frac{5tv}{2r}, \quad G(\xi) = \frac{\rho}{\rho_0}, \quad Z(\xi) = \frac{25t^2c^2}{4r^2}.$$

The conditions at the shock front $\xi = 1$ becomes

$$V(1) = \frac{2}{\gamma+1}, \quad G(1) = \frac{\gamma+1}{\gamma-1}, \quad Z(1) = \frac{2\gamma(\gamma-1)}{(\gamma+1)^2}.$$

Substituting the self-similar variables into the governing equations will lead to three ordinary differential equations. Solving these differential equations analytically is laborious, as shown by Sedov in 1946 and von Neumann in 1947. G. I. Taylor integrated these equations numerically to obtain desired results.